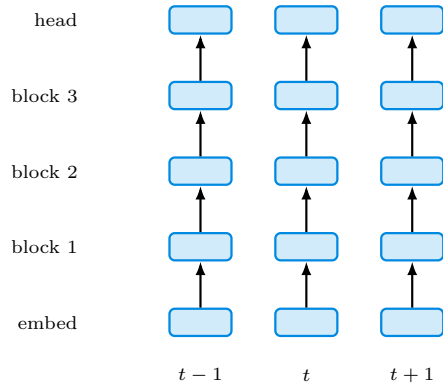


Training-time information flows

Depth recurrence OFF

Vanilla transformer

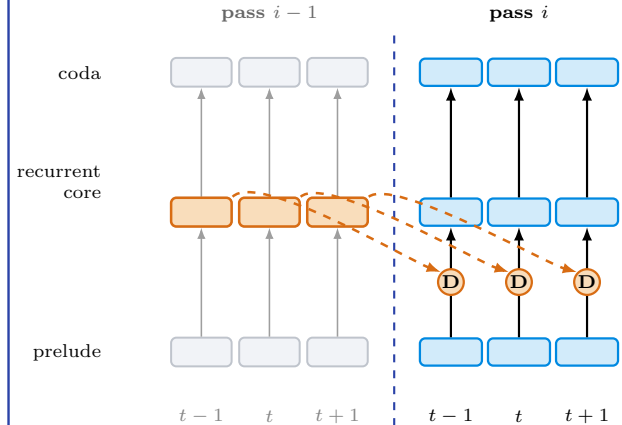
Single pass only. Stages are embedding, backbone blocks, and head.



Depth recurrence ON

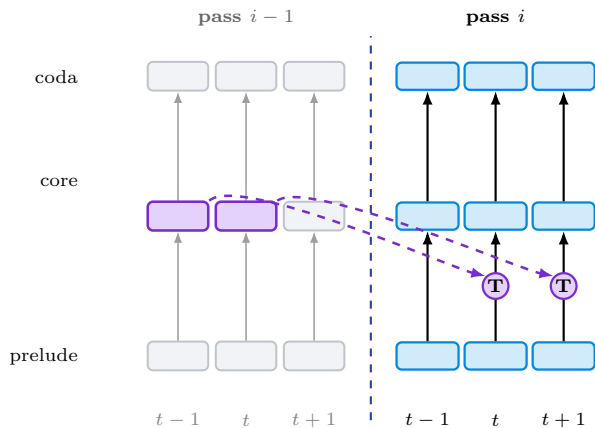
Depth-recurrent / looped LM

Depth state is emitted from recurrent core in pass $i-1$ and read before recurrent core in pass i .



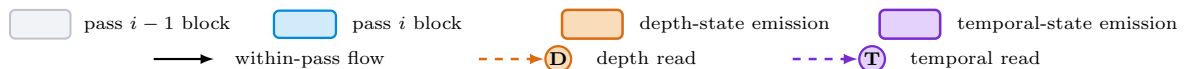
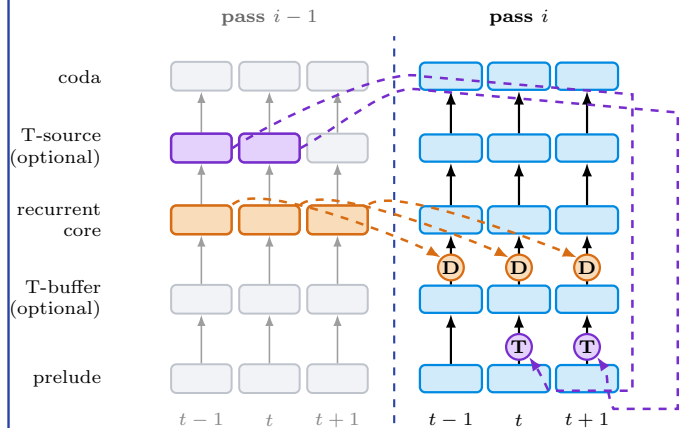
Temporally recurrent

Temporal state is emitted from the last core layer in pass $i-1$ and read before core in pass i .



Depth + temporal hybrid

T-source emits temporal state and recurrent core emits depth state in pass $i-1$; pass i reads them before T-buffer and recurrent core.



Temporal recurrence OFF

Temporal recurrence ON